COHOMOLOGY ON THE BIG ETALE SITE COMPUTES COHOMOLOGY ON THE SMALL ONE, WITH THE SIX COMPARISON IDENTITIES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the K-injective computation for the abelian cases with the composite pullback identity, and the general comparison with the factorisation of the derived pullback for the module cases.

Editorial note. The label lemma-compare-cohomology names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Injectivity and K-injectivity of the restrictions from the big to the small site. Same chapter, the lemma stating that the pullback of an injective or K-injective abelian sheaf on the big site stays injective on a small one. Supplies the computation of part (5) through a K-injective representative.
- Comparison of cohomology along a continuous cocontinuous functor. Cohomology on Sites, the general lemma comparing cohomology on an object with cohomology on its image under such a functor. Supplies part (6) directly.

Source reference. Original reference 0758: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 21102–21102. Prerequisite pointer invoked at proof line(s) 21295; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0DD8: sources/stacks-chow/tags.txt, source lines 16758–16758. Prerequisite pointer invoked at proof line(s) 21302; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : T \to S$ be a morphism of schemes. Then*

(1) *For K in $D(S_{étale})$ we have $H^n_{étale}(S, \pi_S^{-1}K) = H^n(S_{étale}, K)$.*
(2) *For K in $D(S_{étale}, \mathcal{O}_S)$ we have $H^n_{étale}(S, L\pi_S^*K) = H^n(S_{étale}, K)$.*
(3) *For K in $D(S_{étale})$ we have $H^n_{étale}(T, \pi_S^{-1}K) = H^n(T_{étale}, f_{small}^{-1}K)$.*
(4) *For K in $D(S_{étale}, \mathcal{O}_S)$ we have $H^n_{étale}(T, L\pi_S^*K) = H^n(T_{étale}, Lf_{small}^*K)$.*

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(5) *For M in $D((Sch/S)_{étale})$ we have $H^n_{étale}(T, M) = H^n(T_{étale}, i_f^{-1}M)$.*
(6) *For M in $D((Sch/S)_{étale}, \mathcal{O})$ we have $H^n_{étale}(T, M) = H^n(T_{étale}, i_f^*M)$.*

Proof. To prove (5) represent M by a K-injective complex of abelian sheaves and apply Lemma 0758 and work out the definitions. Part (3) follows from this as $i_f^{-1}\pi_S^{-1} = f_{small}^{-1}$. Part (1) is a special case of (3).

Part (6) follows from the very general Cohomology on Sites, Lemma 0DD8. Then part (4) follows because $Lf_{small}^* = i_f^* \circ L\pi_S^*$. Part (2) is a special case of (4). □